

Scylla and Charybdis: a small note on the incompatibility between (certain) statistical and the neurobiological models of human language.

Contemporary linguistics, in particular formal syntax seems to me to be facing a double challenge: on the one hand, the revival of statistics based automated discovery procedures such as those exemplified by (v)LLM; for many, these are considered a valid alternative for understanding language structure and language acquisition to the formal methods established in the second part of the XX century by generative grammar(s) (Chomsky et al. 2023 and references cited there); in the other, neurobiologists, both within the subfield of neurophysiology and evolutionary theories, claim that language structure is the emancipated result of language independent modules, such as those stemming from motor and action planning, obviously common to many other animals (Pulvermüller et al. 2010, Lieberman 2010, Corballis 2003). The perilous rout between the Scylla of statistics and the Charybdis of neurobiologists cannot be avoided if linguistics wants to maintain its autonomous scientific force and euristic value.

Given this scenario, at least two consequences appear to me in need to be highlighted: first, the two approaches are not just not converging: they are in fact totally incompatible, since motor and action planning is not acquired by “statistical intelligence” but it is rather driven and restricted by the anatomical and neurophysiological architecture; second, in neither case is the key-notion of “impossible language” relevant (see Moro 2008 for the original proposal and Moro 2016 for a revised one). Assuming “impossible languages” to be conceivable, coherent and perhaps even simpler grammars which are not recognized by the natural networks exclusively dedicated of language, in fact the following twofold conclusion necessarily emerges: for (v)LLM, impossible languages cannot be defined because there are not such things as natural circuits for machines; the case of action and motor planning, instead, impossible languages cannot be defined because language structure is claimed to be directly linked to the degree of control of movement allowed by the anatomy of the human body and as such if they are conceivable they must also be possible (Moro et al. 2023).

There has been an intense and – at least to me - sometimes ideological and myopic discussion about the role of statistical methods to understand human languages (plural matters here), witness the amount of papers and debates on (v)LLM. But on contrary there has been a relatively quite scarce attention to the fact that the other side of the vision on language has received far less attention, especially when it comes to its inconsistency with the former one. The following two short papers, reproduced here, the former of which has never been available on this platform, clarify why the “action-motor emancipated theories” fail to be plausible models for language and offer theoretical and empirical reason to why

these two approaches are incompatible. It's in between them that the force and the autonomy of linguistics should receive new support to be (re) established as a fruitful domain of scientific inquiry.

References

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